Can Simple Image Preprocessing Recover YOLO Object-Detection Performance After JPEG Compression?

Mini Research #2 — Final Research Paper

Research question: Can simple image preprocessing techniques recover YOLO object-detection performance after JPEG compression?

Dataset: 20 selected COCO train2017 images; 238 non-crowd objects. **Detector:** pretrained Ultralytics YOLO26n.

Abstract

JPEG compression can remove image information that object detectors rely on. This study evaluates whether three simple preprocessing methods can recover detection performance after compression. A pretrained YOLO26n detector was evaluated on 20 selected COCO train2017 images containing 238 non-crowd objects. Original images and JPEG Q70, Q50, and Q30 versions were compared, followed by sharpening, denoising, and 2× Lanczos upscaling applied to Q30. Q30 reduced mAP50 from 0.6166 to 0.5367 (–12.95%), mAP50-95 from 0.4431 to 0.3544 (–20.02%), and Recall from 0.3613 to 0.2983 (–17.44%). Denoising provided the strongest recovery, increasing mAP50-95 to 0.3709 and Recall to 0.3487, but did not restore original performance. Sharpening improved Precision while reducing Recall and mAP, and upscaling produced little benefit.

1. Introduction

JPEG is widely used because it reduces image size through lossy compression. Object detectors may be sensitive to the resulting changes in edges, textures, and object boundaries. YOLO models perform object localization and classification directly from images. This study isolates the effect of JPEG compression and tests whether simple preprocessing can compensate for some lost information. The experiment is intentionally small and controlled and is not intended as a full benchmark.

2. Research Question and Hypotheses

H1: strong JPEG compression reduces detection performance. H2: at least one preprocessing method recovers part of the lost performance. H3: preprocessing methods can affect Precision and Recall differently, so no single metric should define success.

3. Dataset and Experimental Design

The study uses 20 selected COCO train2017 images with 238 non-crowd ground-truth bounding boxes. Phase 1 compares Original, Q70, Q50, and Q30. Phase 2 applies sharpening, denoising, and 2× Lanczos upscaling only to Q30. The same YOLO26n checkpoint and ground-truth annotations are used across conditions.

4. Evaluation

Precision and Recall use confidence ≥ 0.25 and IoU ≥ 0.50. AP is computed by ranking detections by confidence and integrating the precision envelope. mAP50 is macro-averaged across classes present in the subset at IoU 0.50; mAP50-95 averages across IoU thresholds 0.50–0.95. These are study-specific evaluation values, not official COCO benchmark scores.

5. Results

Condition	TP	FP	FN	Precision	Recall	mAP50	mAP50-95
Original	86	22	152	0.7963	0.3613	0.6166	0.4431
Q70	80	23	158	0.7767	0.3361	0.5717	0.4132

Condition	TP	FP	FN	Precision	Recall	mAP50	mAP50-95
Q50	84	23	154	0.7850	0.3529	0.5842	0.4075
Q30	71	17	167	0.8068	0.2983	0.5367	0.3544
Q30+Sharpen	55	10	183	0.8462	0.2311	0.4736	0.2895
Q30+Denoise	83	22	155	0.7905	0.3487	0.5396	0.3709
Q30+Upscale	73	19	165	0.7935	0.3067	0.5324	0.3497

Q30 reduced mAP50 by 12.95%, mAP50-95 by 20.02%, and Recall by 17.44% relative to Original. The slight Precision increase at Q30 does not represent overall improvement because the detector missed more ground-truth objects.

6. Recovery Results

Sharpening raised Precision from 0.8068 to 0.8462 but reduced Recall to 0.2311 and lowered both mAP measures. Denoising was the strongest recovery method, raising Recall to 0.3487 and mAP50-95 to 0.3709. Upscaling produced only small changes and did not recover Q30 performance.

7. Discussion

The experiment indicates that strong JPEG compression can degrade both detection and localization quality. Denoising produced the clearest recovery, especially in Recall, but the improvement was limited. Sharpening demonstrates that visually stronger edges are not necessarily better detector features, while upscaling cannot reconstruct information removed by compression.

8. Limitations

The main limitations are the small 20-image sample, selected COCO subset, single detector, single parameter setting per preprocessing method, single JPEG encoding, and lack of formal statistical significance testing. The reported mAP values are study-specific rather than official COCO benchmark scores.

9. Conclusion

Under the tested conditions, simple preprocessing can recover a limited portion of YOLO detection performance after JPEG compression, but recovery is method-dependent and incomplete. Denoising was the most promising of the three tested methods, while sharpening and upscaling did not provide useful overall recovery.

References

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